

Graphical Abstract

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Highlights

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- Research highlights item 1
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ARTICLE INFO

Keywords:

Porous airfoil
Passive flow control
Panel method
Pressure-driven internal flow
Anisotropic porosity

ABSTRACT

Porous airfoils offer a passive means of modifying aerodynamic loading through pressure-driven flow within internal pathways. This study develops a coupled aerodynamic–hydraulic framework for a NACA 0018 airfoil, combining a source–vortex panel method with rectangular-channel flow models. Three configurations are investigated: chordwise horizontal passages, vertical passages connecting the pressure and suction surfaces, and a combined horizontal–vertical arrangement. The results show that aerodynamic performance depends strongly on pore orientation and connectivity. Horizontal passages redistribute pressure between chordwise locations on the same surface and increase lift at positive angles of attack, with stronger effects at larger pore heights. Vertical pores promote pressure equalisation between the pressure and suction surfaces and reduce lift relative to the impermeable airfoil. In the combined configuration, these mechanisms interact through the external flow field, giving a response closer to that of the solid airfoil. Pressure and velocity difference fields show that the strongest flow modifications occur near the pore openings. These results demonstrate that porous-airfoil performance is governed not simply by porosity, but by the orientation, location and hydraulic connectivity of the internal passage

1. Introduction

Improving airfoil performance without relying on powered control systems remains an important objective in aerospace and energy applications. Conventional high-lift and flow-control devices can be effective, but they may introduce mechanical complexity, additional mass, maintenance requirements, and changes that disrupt the smooth external surface of the airfoil. Passive flow-control concepts provide an alternative strategy in which the pressure field surrounding an aerodynamic body supplies the energy required to modify the flow. Porous surfaces, permeable regions, localised pores, and internal passages employ this principle by permitting fluid transport between regions of unequal pressure. Their effectiveness, however, depends on the placement of the pores, the manner in which they are connected, and the strength of the interaction between the resulting surface normal flow and the external aerodynamic field.

Early investigations established that surface porosity can substantially modify the pressure distribution and aerodynamic forces acting on an airfoil, while also demonstrating that porosity is not inherently beneficial. Mineck and Hartwich examined full-chord porosity on a NACA 0012 airfoil under transonic-flow conditions and found that the porous surface generally reduced the rate at which lift increased with angle of attack and increased drag for a given normal-force coefficient [1]. Experiments by Aldheeb et al. similarly showed that increasing porosity reduced the rate of lift increase. They also found that a small amount of porosity

initially reduced drag, whereas further increases in porosity caused drag to rise [2]. Together, these findings show that allowing pressure to equalise freely across the airfoil can reduce the pressure difference required for lift generation and may adversely affect aerodynamic performance. Effective passive flow control therefore requires selective permeability or controlled connectivity rather than indiscriminate porosity across the airfoil.

Reduced-order models have provided important insight into how distributed permeability affects airfoil aerodynamics. Hajian and Jaworski introduced a steady thin-airfoil formulation in which a local Darcy condition relates the pressure difference across the airfoil to the flow through a chordwise distribution of porosity [3]. Luesutthiviboon et al. assessed a related thin-airfoil model for partially porous airfoils and found that its accuracy decreased when the porous region was small or the flow resistance was low [4]. Baddoo et al. subsequently extended porous-airfoil modelling to unsteady motion and gust response by including the effects of flow resistance and the inertia of the fluid moving through the porous surface [5]. These models are computationally efficient and explain the relationship between pressure difference, flow resistance, and the resulting aerodynamic forces. However, thin-airfoil formulations do not represent the finite thickness of an airfoil. Their porous boundary conditions also primarily describe local flow between the pressure and suction surfaces through distributed or homogenised porous material, rather than flow through finite-length internal passages connecting selected and spatially separated surface pores.

Experiments and high-fidelity simulations also show that the internal structure and directionality of the permeable region are important. Hanna and Spedding studied a NACA 0012 wing containing discrete wall-normal holes and observed that a small degree of porosity modified

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the lift response around zero incidence while maintaining comparable lift-to-drag performance within the experimental uncertainty over much of the tested range [6]. For a NACA 0018 airfoil, Zhu et al. compared fully porous and suction-side porous trailing-edge arrangements and found distinct aerodynamic and acoustic responses, demonstrating the sensitivity of the outcome to porous placement and cross-surface pressure release [7]. Li et al. compared an isotropic porous medium with directionally constrained straight holes using large-eddy simulation and showed that aerodynamic behaviour depends not only on porous extent and porosity but also on the internal flow direction permitted by the structure [8]. Taken together, these studies show that a bulk porosity or permeability value alone is insufficient to describe the aerodynamic function of a designed internal architecture.

An alternative class of passive devices uses internal slots or channels to transport fluid from a region of high pressure to one of lower pressure. Acarer optimised passive slots for the DU12W262 airfoil and reported an improvement in the peak lift-to-drag ratio, demonstrating the potential of pressure-driven flow through internal channels to improve aerodynamic performance [9]. Kestel et al. used passive flow through internal channels to modify the separated vortex structure over a non-slender delta wing, directly demonstrating fluid transport driven by the natural pressure difference between connected surfaces [10]. More recently, Bahadi et al. conducted a parametric investigation of a curved internal slot in a NACA 4412 airfoil and reported increased aerodynamic performance at high angles of attack for selected inlet, taper, and curvature configurations [11]. These studies demonstrate that internal pressure-driven connections have been investigated previously. Therefore, the contribution of the present study is not the initial introduction of passive connectivity, but the development of a general reduced-order framework for modelling multiple pairs of pores connected by independent internal channels. Previous investigations have generally focused on a specific slot geometry using experiments or Reynolds-averaged Navier–Stokes calculations.

Porous-airfoil design is increasingly being treated as an optimisation problem because aerodynamic performance depends on several interacting geometric and material variables. Zamani et al. applied a parameter-screening and optimisation procedure to a porous NACA 0015 airfoil for static-stall control, varying the dimensions and location of the porous region, together with its porosity and cell size [12]. This work demonstrates the value of systematic design exploration, although the porous layer is represented using distributed material properties. For airfoils with individual internal channels, the design variables include which surface pores are connected, where the pores are positioned, the length, orientation, and height of each channel, and whether the channels connect pores on the same surface or link the pressure and suction surfaces. Exploring this larger range of possible channel arrangements exclusively through viscous

computational fluid dynamics or experiments would be computationally expensive during the preliminary design stage.

Panel methods bridge the gap between simplified analytical thin-airfoil models and computationally expensive viscous-flow simulations. The foundational work of Hess and Smith established a boundary-element representation of potential flow by distributing singularities along the boundary of the geometry. This approach enables the pressure field around a geometrically resolved body to be calculated without discretising the surrounding fluid domain [13]. The subsequent review by Hess documented the development, capabilities, and limitations of panel methods as efficient aerodynamic boundary-integral techniques [14], while Birk provided a modern, implementation-focused description of the Hess–Smith method using constant-strength source and doublet distributions [15]. For a two-dimensional lifting airfoil, a source distribution enforces the surface-normal boundary condition, while circulation and a trailing-edge Kutta condition produce the lifting solution. This structure is particularly suitable for the present problem because the standard impermeability condition can be modified locally by prescribing a normal flow velocity at each surface pore. The calculated surface velocity determines the pressure coefficient, from which the pressure difference between each pair of connected pores is obtained to drive the internal channel flow. Panel methods are also computationally inexpensive, allowing repeated calculations for numerous channel arrangements, pore sizes, and angles of attack.

Established viscous–inviscid methods show how potential-flow panel calculations can be combined with boundary-layer models to provide a more complete analysis of airfoil performance. Drela and Giles developed a strongly coupled viscous–inviscid method for transonic and low-Reynolds-number airfoils by combining an inviscid outer-flow solution with integral boundary-layer equations [16]. XFOIL later implemented a similar approach in a computationally efficient airfoil-analysis and design tool, using a linear-vorticity panel method for the inviscid flow together with integral boundary-layer equations and a transition-prediction model [17]. Although these methods predict viscous effects that are not included in the present formulation, their use of panel methods for the external flow supports the efficient calculation of surface-pressure distributions using potential-flow theory. In the present study, XFOIL is used only as a reference for the solid-airfoil baseline. The proposed external-flow solver remains inviscid so that the pressure-driven aero–hydraulic interaction can be isolated and evaluated at a low computational cost. This modelling choice also makes the method suitable for preliminary design screening, after which selected channel configurations can be investigated using viscous computational fluid dynamics or experiments.

The reviewed literature therefore covers distributed Darcy-type surfaces, homogenised porous media, pores oriented normal to the airfoil surface, individual passive slots, and established potential-flow panel formulations. However, a modelling gap remains between local permeability boundary conditions and high-fidelity simulations of specific internal

geometries. In particular, limited attention has been given to a panel-based reduced-order framework that explicitly connects selected pairs of surface pores using finite-length internal channels, calculates the pressure-driven flow through each channel, and uses the resulting inflow and outflow velocities at the pores to update the external aerodynamic solution. The systematic comparison of channels connecting spatially separated pores on the same surface, channels connecting the pressure and suction surfaces, and a combined arrangement containing both channel types as hydraulically independent connections also remains underexplored. Addressing this gap would provide a computationally efficient method for screening many candidate channel layouts before conducting higher-fidelity viscous simulations and experimental validation.

Accordingly, the present study develops a two-dimensional coupled aero-hydraulic framework for a NACA 0018 airfoil. The external flow is represented by a steady, incompressible, inviscid, and irrotational source-vortex panel method with a Kutta condition. Selected pairs of surface pores are connected by passive rectangular channels, and the pressure difference between the paired pores determines the internal flow according to a hydraulic-resistance relation. The resulting surface-normal flow velocities at the pores are iteratively coupled with the external panel solution. Three channel arrangements are examined: four chordwise channels connecting pores on the same surface, four vertical channels connecting the pressure and suction surfaces, and a combined arrangement containing all eight channels as hydraulically independent connections. Channel heights of 8, 9, and 10 mm are assessed over angles of attack from -5° to 15° at a chord Reynolds number of 5×10^5 . The results are compared with those of the solid airfoil and an XFOIL reference solution.

The main contribution of this study is a reduced-order representation of finite-length internal channels connecting selected pairs of surface pores within a coupled external-flow and internal-channel model. The formulation allows different channel arrangements to be evaluated using a consistent aerodynamic model and demonstrates the relationships among the pressure difference between connected pores, internal flow rate, surface-normal flow velocity, pressure distribution, lift, pitching moment, and pressure-based streamwise force. The method is intended for rapid preliminary assessment and comparison of candidate channel layouts. Because the external-flow model does not resolve boundary layers, skin-friction drag, flow separation, or stall, the results are interpreted as inviscid aerodynamic trends and used to select promising configurations for subsequent viscous computational analysis and experimental validation.

2. Methods

2.1. Modelling framework and assumptions

A two-dimensional coupled aero-hydraulic model was developed for a NACA 0018 airfoil containing passive internal channels. In this study, a *pore* denotes the finite surface

region at which a channel intersects the airfoil boundary, whereas a *channel* denotes the finite-length internal connection between two pores. The external flow is represented by a steady source-vortex panel method, and the flow in each channel is represented by a one-dimensional pressure-loss model per unit span. The two models exchange surface pressure and pore-normal velocity until their flow rates are mutually consistent.

The external model assumes incompressible, inviscid and irrotational flow. It therefore does not resolve boundary layers, skin friction, separation, transition or stall. The internal model assumes a straight, smooth, constant-height rectangular channel described by its cross-sectionally averaged velocity. Entrance turning, three-dimensional pore-mouth flow and jet mixing are not resolved explicitly; their treatment in the reduced energy closure is stated in Section 2.4. Accordingly, the method is intended for rapid comparison and screening of channel layouts rather than as a substitute for viscous computational-fluid-dynamics or experimental analysis.

The first flow estimate is calculated from the solid-airfoil pressure distribution, with zero surface-normal velocity. The pressure at each pore is supplied to the channel model to obtain an initial channel flow rate. These flows produce suction and blowing at the pores, which change the external pressures. The channel flow rates are therefore adjusted until they agree with the flow rates driven by the updated pressures. A nearby converged solution is used as the initial estimate for subsequent cases where available. Figure 1 summarises the calculation.

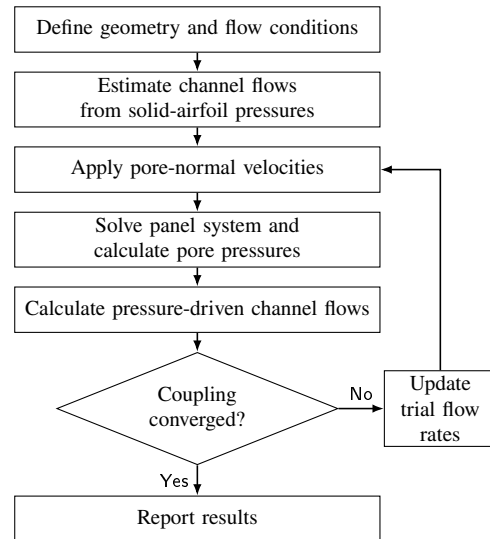


Figure 1: Coupled aero-hydraulic solution procedure.

2.2. Airfoil and channel geometry

The closed NACA four-digit profile is generated from the standard analytical ordinates [18]. The coefficient -0.1036 is used for the fourth-order thickness term to obtain a closed trailing edge. Cosine-spaced points are placed on each surface and joined by N straight panels. Panel i has length S_i ,

control point (x_i, y_i) , unit tangent $\mathbf{t}_i = (t_{x,i}, t_{y,i})$ and outward unit normal $\mathbf{n}_i = (n_{x,i}, n_{y,i})$.

Three layouts are considered:

1. **Chordwise layout:** four independent same-surface channels, comprising two on the upper surface and two on the lower surface;
2. **Cross-surface layout:** four independent channels connecting the lower and upper surfaces at chordwise positions distributed from approximately $x/c = 0.10$ to 0.89 ; and
3. **Combined layout:** all eight channels from the preceding layouts, retained as hydraulically independent connections with no internal junction or exchange of flow between channels.

The layouts are shown in Figure 2. A crossing of projected channel centrelines in the two-dimensional drawing does not create a hydraulic connection. Such crossings represent an idealised independent-channel model, not a resolved, physically intersecting two-dimensional channel geometry; a realisation would require separation of the channels. Surface pores are not permitted to overlap.

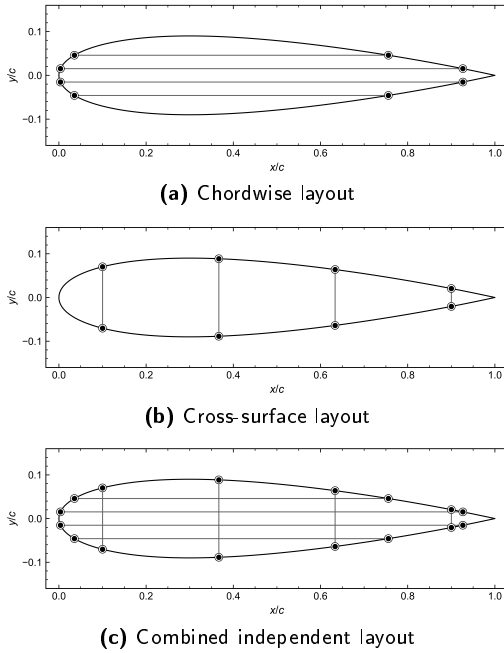


Figure 2: Internal-channel layouts considered in this study.

Each pore is specified by a chordwise fraction, a surface identifier and the channel height H_k . The requested location is mapped to the polygonal airfoil surface independently of the panel control points. The channel centreline length L_k is the straight-line distance between its two mapped surface endpoints. Before solution, the geometry is checked to ensure that every pore lies within the permitted chordwise range, that different pores do not overlap, that the specified minimum separation is maintained and that the channel fits within the local airfoil thickness. A geometry that fails any check is classified as invalid and is not included as a converged aerodynamic case.

2.3. External aerodynamic model

2.3.1. Source-vortex panel formulation

The panel method follows the boundary-integral representation of incompressible potential flow developed by Hess and Smith and described in later panel-method treatments [13, 19, 15]. With

$$\mathbf{V} = \nabla \Phi, \quad \nabla^2 \Phi = 0, \quad (1)$$

the external velocity is represented by the freestream, one constant source strength λ_j on every panel and one uniform vortex-sheet strength γ over the closed boundary. The sources represent thickness, while the vortex sheet provides the circulation required for a lifting solution.

For a solid surface, the normal velocity at every control point is zero. At a pore, this boundary condition is replaced by a prescribed normal transpiration velocity, an approach consistent with reduced porous-surface formulations [3, 5]:

$$\mathbf{V}_i \cdot \mathbf{n}_i = v_{n,i}, \quad (2)$$

where $v_{n,i} > 0$ denotes flow out of the airfoil and $v_{n,i} < 0$ denotes flow into it. With the adopted influence-coefficient convention, the discrete normal equation is

$$\pi \lambda_i + \sum_{\substack{j=1 \\ j \neq i}}^N I_{ij} \lambda_j - \gamma \sum_{j=1}^N K_{ij} = -2\pi V_\infty \cos \beta_i + 2\pi v_{n,i}, \quad (3)$$

where I_{ij} and K_{ij} are the normal source and vortex influence coefficients and β_i is the angle between the panel normal and the freestream direction. The trailing-edge Kutta condition is

$$V_{t,1} + V_{t,N} = 0. \quad (4)$$

Equations (3) and (4) form an $(N+1) \times (N+1)$ linear system for $[\lambda_1, \dots, \lambda_N, \gamma]^T$. Because the influence matrix depends only on geometry, it is assembled and LU-factorised once for each airfoil discretisation; only the right-hand side changes when α or v_n changes.

2.3.2. Surface velocity and pressure

After solution of the panel system, the tangential velocity on panel i is

$$V_{t,i} = V_\infty \sin \beta_i + \frac{1}{2\pi} \sum_{j=1}^N \lambda_j J_{ij} + \frac{\gamma}{2} - \frac{\gamma}{2\pi} \sum_{j=1}^N L_{ij}, \quad (5)$$

where J_{ij} and L_{ij} are the tangential source and vortex influence coefficients. The prescribed normal component is retained when Bernoulli's equation is evaluated at a pore:

$$C_{p,i} = 1 - \frac{V_{t,i}^2 + v_{n,i}^2}{V_\infty^2}, \quad p_i = p_\infty + \frac{1}{2} \rho_\infty V_\infty^2 C_{p,i}. \quad (6)$$

This expression is a potential-flow closure at the exterior surface; it does not resolve the dissipative pore-mouth region.

2.4. Internal rectangular-channel model

Channel k is treated as a two-dimensional rectangular duct of height H_k and length L_k , with all flow quantities expressed per unit span. The signed flow rate per unit span q'_k , positive from labelled pore 1 to labelled pore 2, and bulk velocity U_k are related by

$$U_k = \frac{q'_k}{H_k}, \quad D_{h,k} = 2H_k, \quad Re_{h,k} = \frac{\rho_\infty |U_k| D_{h,k}}{\mu_\infty}. \quad (7)$$

The Darcy–Weisbach friction loss is

$$\Delta p_{f,k} = f_k \frac{L_k}{D_{h,k}} \frac{\rho_\infty U_k^2}{2}. \quad (8)$$

For $Re_{h,k} \leq 2000$, the parallel-wall result $f_k = 96/Re_{h,k}$ is used [20]; substitution into Eq. (8) gives the plane-Poiseuille limit

$$\Delta p_{f,k} = \frac{12\mu_\infty L_k}{H_k^2} |U_k|, \quad R'_k = \frac{\Delta p_{f,k}}{|q'_k|} = \frac{12\mu_\infty L_k}{H_k^3}. \quad (9)$$

For $Re_{h,k} \geq 3000$, the adopted smooth-wall approximation is $f_k = 0.3164 Re_{h,k}^{-1/4}$. This pipe-flow correlation is extended to the parallel-wall channel using its hydraulic diameter and is not a geometry-specific turbulent-channel validation. The Darcy friction-factor convention is used throughout [21]. A continuously differentiable blend is applied between the two limits:

$$\xi_k = \frac{Re_{h,k} - 2000}{1000}, \quad w_k = \xi_k^2 (3 - 2\xi_k),$$

$$f_k = (1 - w_k) \frac{96}{Re_{h,k}} + w_k \frac{0.3164}{Re_{h,k}^{1/4}}. \quad (10)$$

The blended relation is an engineering closure used to avoid a discontinuity; it is not a transition model for the developing pore flow.

Let $p_{in,k}$ and $p_{out,k}$ denote the area-averaged static pressures at the higher- and lower-pressure pores, respectively. The active energy closure is

$$p_{in,k} - p_{out,k} + \left\langle \frac{1}{2} \rho_\infty v_n^2 \right\rangle_{in,k} = \Delta p_{f,k} + \alpha_j \frac{1}{2} \rho_\infty U_k^2, \quad (11)$$

with $\alpha_j = 1$. The external tangential kinetic head is assumed to be lost during turning into the channel and is therefore not credited as available channel head. At the outlet, the static pressure is matched to the exterior, while the bulk jet kinetic head is carried out of the channel. Equation (11) is monotonic in $|U_k|$ and is solved by a bracketed scalar bisection. The flow direction is taken from the sign of the static pore-pressure difference; an exact pressure tie gives zero flow. This closure assumes zero recovery of inlet tangential kinetic energy; it has not been calibrated against a resolved three-dimensional pore-mouth flow and should be interpreted as a reduced-order modelling assumption.

2.5. Pore averaging and aero–hydraulic coupling

2.5.1. Pressure averaging and flow transfer

The intersection of a channel strip with the polygonal surface defines the finite extent of each pore. Surface pressure is reconstructed piecewise linearly through panel-centre values and integrated along the pore. Thus, for pore m of channel k ,

$$p_{m,k} = \frac{1}{\ell_{m,k}} \int_{\mathcal{P}_{m,k}} p(s) ds, \quad \ell_{m,k} = \int_{\mathcal{P}_{m,k}} ds. \quad (12)$$

This treatment allows a pore boundary to cut through a panel without moving the prescribed physical location.

The channel flow is transferred to the intersected panels as a panel-average normal velocity. If $\ell_{i,m,k}$ is the portion of panel i covered by pore m of channel k , then

$$v_{n,i}^{(m,k)} = \sigma_{m,k} \frac{q'_k \ell_{i,m,k}}{\ell_{m,k} S_i}, \quad \sigma_{m,k} = \begin{cases} -1, & m = 1, \\ +1, & m = 2. \end{cases} \quad (13)$$

Consequently,

$$\sum_{i \in \mathcal{P}_{m,k}} v_{n,i}^{(m,k)} S_i = \sigma_{m,k} q'_k, \quad (14)$$

For $q'_k > 0$, pore 1 is the inlet and pore 2 is the outlet; these roles reverse when $q'_k < 0$. Thus each independently connected channel conserves flow exactly within the two-dimensional per-unit-span formulation. Contributions are summed if more than one channel acts on the aerodynamic boundary, although overlapping pores are excluded by the geometry checks.

2.5.2. Nonlinearity and solution procedure

The panel equations are linear when the pore-normal velocities are prescribed. The coupled problem is nonlinear because these velocities depend on channel flow rates driven by the pressure distribution they produce. Pressure depends on squared velocity through Bernoulli's equation, while the channel model includes a Reynolds-number-dependent friction factor and an outlet kinetic-head term. Directly repeating pressure and flow updates can therefore oscillate or fail to converge. Larger channels can strengthen this feedback because the laminar hydraulic resistance decreases as H_k^{-3} .

To obtain a consistent solution, all channel flow rates are solved together as a nonlinear root problem. The dimensionless unknowns are

$$\hat{q}_k = \frac{q'_k}{H_k V_\infty}. \quad (15)$$

For a trial vector, the panel solver and channel model return the pressure-driven flow vector $\hat{q}'_{hyd}$. The residual is

$$R_k(\hat{q}) = \frac{q'_{k,trial} - q'_{k,hyd}}{H_k V_\infty}. \quad (16)$$

A Powell hybrid root solver adjusts the trial flows to reduce this mismatch, with a trust-region reflective least-squares

method used if required [22, 23]. Each trial requires a linear panel solve followed by the channel-flow calculation. A solution is accepted only when the scaled residual satisfies

$$\|\mathbf{R}\|_\infty \leq \sqrt{\epsilon_{\text{mach}}} \max(1, \|\hat{\mathbf{q}}\|_\infty). \quad (17)$$

where ϵ_{mach} is machine precision. A successful solver termination alone is insufficient if this condition is not met.

The angle-of-attack sweep begins at the requested angle nearest to zero and advances towards positive and negative angles, using converged channel flows to initialise neighbouring cases. For difficult cases, continuation introduces intermediate operating conditions or gradually increases the coupling strength. The available strategies include adaptive angle steps, Reynolds-number continuation where applicable, coupling homotopy and pseudo-arclength continuation. Only solutions satisfying the fully coupled equations at the requested operating conditions are retained. Invalid geometries are rejected before solution, and cases that remain unconverged after the available attempts are reported as NaN. Non-convergence does not establish stall or prove that no steady solution exists.

2.6. Aerodynamic output coefficients

Pressure forces are integrated over the closed polygonal boundary:

$$\begin{aligned} dC_{x,i} &= -C_{p,i} n_{x,i} \frac{S_i}{c}, & dC_{y,i} &= -C_{p,i} n_{y,i} \frac{S_i}{c}, \\ C_x &= \sum_{i=1}^N dC_{x,i}, & C_y &= \sum_{i=1}^N dC_{y,i}. \end{aligned} \quad (18)$$

The freestream-aligned coefficients are

$$\begin{aligned} C_L &= C_y \cos \alpha - C_x \sin \alpha, \\ C_{D,p} &= C_x \cos \alpha + C_y \sin \alpha. \end{aligned} \quad (19)$$

Here $C_{D,p}$ is a closed-contour pressure-based streamwise-force coefficient; it is not total drag because skin friction and viscous wake momentum loss are not included. In addition, the integration retains pressure on the nominal contour across the pores and does not explicitly include internal-wall forces or the momentum transported through the pores. It must therefore not be identified with the net force on the complete porous device. The same qualification applies to the pressure-integrated lift and moment. The quarter-chord pitching-moment coefficient is

$$C_M = - \sum_{i=1}^N \frac{(x_i - 0.25c) dC_{y,i} - y_i dC_{x,i}}{c}. \quad (20)$$

As an internal check, a circulation-based lift coefficient is also evaluated:

$$C_{L,\Gamma} = \frac{2\gamma \sum_{i=1}^N S_i}{V_\infty c}. \quad (21)$$

Table 1

Numerical and operating conditions used in the present study.

Parameter	Symbol	Value
Airfoil	–	NACA 0018
Chord length	c	1 m
Number of panels	N	1000
Fluid density	ρ_∞	1.225 kg m ⁻³
Dynamic viscosity	μ_∞	1.8 × 10 ⁻⁵ Pa s
Freestream pressure	p_∞	101325 Pa
Chord Reynolds number	Re_c	5.0 × 10 ⁵
Principal angle of attack	α	4°
Angle-of-attack range	α	–5° to 15°
Angle increment	$\Delta\alpha$	1°
Channel height	H	8, 9, 10 mm

2.7. Numerical conditions and verification

The conditions used for the reported study are listed in Table 1. The freestream speed is calculated from

$$V_\infty = \frac{Re_c \mu_\infty}{\rho_\infty c}. \quad (22)$$

The impermeable airfoil is solved with the same geometry, panel distribution and operating conditions as the channelled configurations. An inviscid XFOIL solution, based on a linear-vorticity panel method, is used as an independent reference for the impermeable-airfoil surface-pressure distribution and angle-of-attack trend [17]. This comparison is a verification of the external-flow implementation and does not validate the internal channel or pore-mouth closures.

A panel-refinement assessment is required for the impermeable airfoil and representative coupled cases, monitoring C_p , C_L , $C_{D,p}$, C_M and channel flow rates. The nominal production value is $N = 1000$; convergence of integrated forces alone does not establish convergence of local pore pressures. For every coupled case, available diagnostics include the panel-system residual, Kutta residual, normal-boundary-condition residual, channel energy residual, maximum $|v_n|/V_\infty$, hydraulic Reynolds number and nonlinear coupling residual. These checks distinguish numerical convergence from the physical limitations of the reduced-order model.

3. Results and discussion

The aerodynamic behaviour of the three porous-airfoil configurations was investigated over an angle-of-attack range of –5° to 15°. Three pore heights, $H = 8, 9$ and 10 mm, were considered for the horizontal, vertical and horizontal–vertical porous airfoils. The corresponding impermeable (solid) airfoil panel-method solution and the inviscid XFOIL prediction are included as reference cases.

The effect of the pores is first examined through the lift and moment coefficients over the angle-of-attack range. The observed aerodynamic trends are then interpreted using the corresponding pressure and velocity field differences. A detailed flow-field comparison is presented at $\alpha = 10^\circ$ for a pore height of $H = 9$ mm.

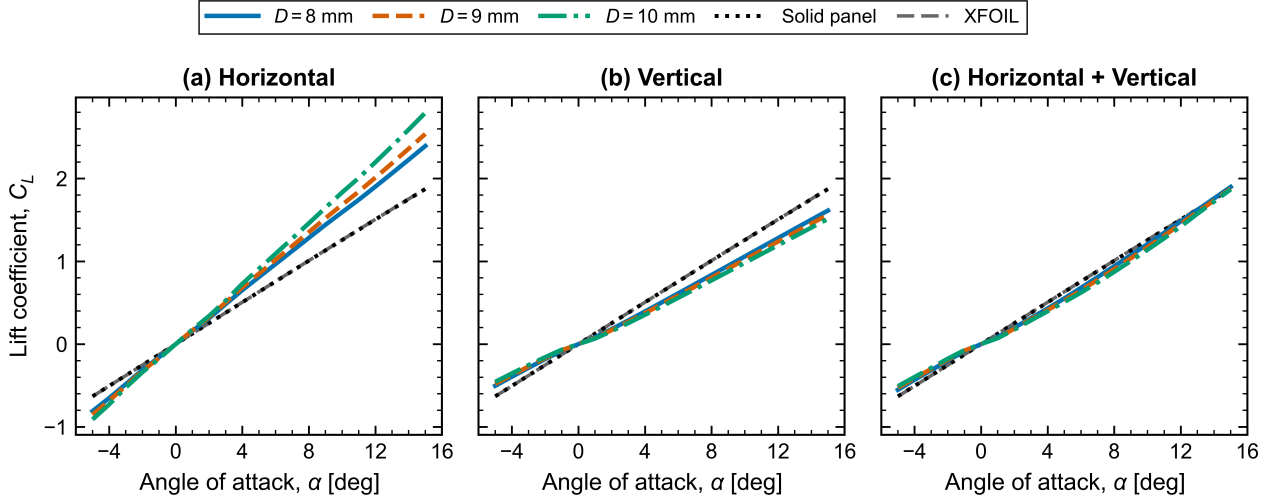


Figure 3: Lift coefficient as a function of angle of attack for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Results are shown for pore heights $H = 8, 9$ and 10 mm together with the impermeable panel-method and inviscid XFOIL reference solutions.

3.1. Panel method validation against XFOIL

The impermeable source–vortex panel-method solution is compared with the corresponding inviscid XFOIL prediction in Figures. 3 and 4. Both methods show similar trends in lift and pitching moment over the investigated angle-of-attack range.

This agreement provides a useful reference for assessing the aerodynamic effects introduced by the pores. Small differences between the panel-method and XFOIL predictions are expected because the two approaches use different numerical formulations, although both represent inviscid external flow.

The impermeable panel-method solution is therefore used as the primary reference when evaluating the porous-airfoil configurations. This ensures that the porous and solid airfoils are compared using the same geometry, panel discretisation, aerodynamic formulation and operating conditions.

3.2. Effect of pores on lift

Figure 3 compares the variation of lift coefficient with angle of attack for the three porous-airfoil configurations and pore heights. The horizontal porous airfoil produces the largest change in lift. At positive angles of attack, its lift coefficient is higher than that of the impermeable reference, and the difference becomes more pronounced as the angle of attack increases. The effect also strengthens with pore height, with $H = 10$ mm producing the highest lift at larger positive angles of attack. At negative angles of attack, increasing the pore height similarly increases the magnitude of the negative lift. This nearly symmetric lift response about the zero-lift condition is consistent with the symmetric NACA 0018 geometry.

For the horizontal configuration, both ends of each pore are located on the same airfoil surface. The streamwise variation in surface pressure therefore creates a pressure

difference between the two ends and drives internal flow from the locally higher-pressure region towards the lower-pressure region. This produces suction at one end and blowing at the other, changing the local surface-normal velocity and redistributing the external pressure and velocity fields along the airfoil surface. At positive angle of attack, the different pressure distributions over the suction and pressure surfaces lead to a net redistribution of aerodynamic loading that increases lift.

The stronger dependence on pore height is consistent with the reduction in hydraulic resistance as the pore becomes larger. For a given pressure difference, a larger pore height permits a greater internal flow and therefore a stronger interaction between the pore flow and the external aerodynamic field.

The vertical porous airfoil shows a distinctly different behaviour. At positive angles of attack, its lift coefficient is generally lower than that of the impermeable airfoil, while the three pore-height cases remain relatively close over most of the investigated range.

In this configuration, each pore connects the pressure surface to the suction surface. The resulting pressure difference drives internal flow between the two surfaces and partially reduces the pressure difference responsible for lift. This pressure equalisation is consistent with the lower lift coefficient obtained for the vertical porous airfoil. The small separation between the pore-height cases also indicates that this configuration is less sensitive to pore height than the horizontal arrangement.

The horizontal–vertical porous airfoil remains comparatively close to the impermeable reference. At positive angles of attack, its lift coefficient is generally slightly lower than that of the solid airfoil, and the three pore-height cases remain closely grouped.

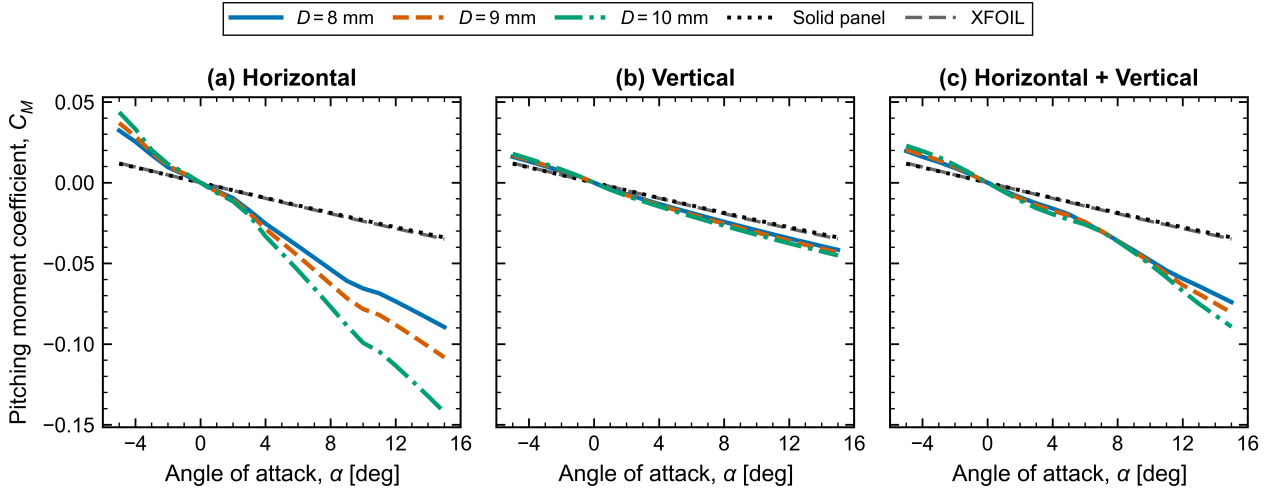


Figure 4: Pitching-moment coefficient about the quarter chord as a function of angle of attack for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal-vertical porous airfoil. Results are shown for pore heights $H = 8, 9$ and 10 mm together with the impermeable panel-method and inviscid XFOIL reference solutions.

This behaviour reflects the simultaneous influence of the horizontal and vertical pore arrangements. The horizontal pores redistribute flow between different chordwise locations on the same surface, whereas the vertical pores transfer flow between the pressure and suction surfaces. The lift-enhancing redistribution associated with the horizontal pores is therefore partly offset by the pressure equalisation introduced by the vertical pores.

The horizontal and vertical pores remain physically separate and hydraulically independent. However, their aerodynamic effects interact through the external pressure field, since changes produced by one set of pores alter the surface pressures that drive the other set. Consequently, the response of the horizontal-vertical porous airfoil is not a simple superposition of the two individual configurations.

Overall, the results show that the aerodynamic effect of the pores is governed mainly by their orientation, placement and the pressure difference between the connected surface locations. Pore height also influences the response, but its effect is strongest for the horizontal configuration and considerably weaker for the vertical and horizontal-vertical arrangements.

3.3. Effect of pores on pitching moment

The corresponding moment coefficients are presented in Figure. 4. The horizontal porous airfoil produces the largest increase from the impermeable-airfoil response. As the positive angle of attack increases, the pitching-moment coefficient becomes progressively more negative. The magnitude of this change also increases with pore height, with $H = 10$ mm producing the largest negative pitching moment at higher positive angles of attack.

This behaviour is associated with the chordwise redistribution of surface pressure caused by the horizontal pores. Since the two ends of each pore are located at different

chordwise positions along a given airfoil surface, the pressure changes induced by the internal flow act at different distances from the quarter-chord reference point. These local pressure variations therefore contribute differently to the pitching moment. Increasing the pore height strengthens the internal flow and the resulting pressure redistribution, leading to a larger change in the pitching-moment coefficient.

The increase in lift obtained with the horizontal porous airfoil is therefore accompanied by a larger negative pitching moment. This represents an important aerodynamic trade-off, since the configuration that provides the greatest lift enhancement also produces the largest change in longitudinal loading.

The vertical porous airfoil produces considerably smaller changes in moment, and the three pore-height cases remain closely grouped over most of the investigated angle-of-attack range. This behaviour is consistent with the weaker dependence on pore height observed in the lift results.

For the vertical configuration, the two ends of each pore are positioned on the upper and lower surfaces at approximately the same chordwise location. The resulting pressure redistribution therefore occurs mainly between the pressure and suction surfaces rather than between widely separated positions along the chord. The corresponding change in the chordwise pressure distribution is smaller than for the horizontal pores, resulting in a weaker pitching-moment response.

The horizontal-vertical porous airfoil exhibits behaviour between that of the individual horizontal and vertical configurations. The pitching moment becomes increasingly negative with positive angle of attack, although the departure from the impermeable-airfoil solution remains smaller than for the horizontal porous airfoil. The influence of pore height also becomes more noticeable at higher positive angles of attack.

In this configuration, the horizontal pores redistribute loading between different chordwise positions, while the vertical pores mainly redistribute pressure between the upper and lower surfaces. The two pore arrangements remain physically separate and hydraulically independent, but their aerodynamic effects interact through the external pressure field. The resulting pitching-moment variation therefore reflects the combined influence of these two pressure-redistribution mechanisms.

Overall, the pitching-moment results show that the effect of the pores depends not only on the extent of the pressure redistribution, but also on where it occurs along the chord. The horizontal porous airfoil produces the largest change in moment because the associated pressure variations occur at widely separated chordwise locations, creating a stronger moment about the quarter-chord reference point. In contrast, the vertical porous airfoil produces a smaller change because its pressure redistribution occurs mainly between the upper and lower surfaces at similar chordwise positions.

3.4. Flow field characteristics of porous airfoils

To examine how the pores alter the external flow, the pressure and velocity fields of the porous airfoils are compared with those of the corresponding impermeable airfoil at $\alpha = 10^\circ$ and $H = 9$ mm.

The comparison is presented using pressure- and velocity-difference contours rather than the absolute flow fields. The absolute contours are dominated by the overall aerodynamic behaviour of the airfoil and therefore make the local influence of the pores more difficult to identify. Subtracting the impermeable-airfoil solution isolates the changes introduced by the pore flow, allowing the location, direction and relative strength of the resulting pressure and velocity variations to be examined more clearly.

3.4.1. Pressure field characteristics

Figure 5 presents the pressure difference between the porous and impermeable airfoils,

$$\Delta p = p_{\text{porous}} - p_{\text{solid}}, \quad (23)$$

for the three pore arrangements.

The pressure difference is shown instead of the absolute pressure field to isolate the changes introduced by the pores. In the absolute solution, the pressure distribution is dominated by the overall airfoil loading and the strong pressure gradients near the leading edge. Subtracting the impermeable-airfoil solution makes the local changes associated with the pore flow easier to identify.

The strongest pressure differences occur near the pore ends, where the internal flow directly influences the surrounding external flow. In several locations, neighbouring positive and negative values of Δp form a dipole-like pattern. This is associated with suction at one end of a pore and blowing at the other, producing local regions of reduced and increased pressure relative to the impermeable airfoil.

The effect of these pressure changes on lift depends on where they occur on the airfoil. At positive angle of attack, a reduction in pressure on the upper suction surface strengthens the suction and contributes to higher lift, whereas an increase in pressure has the opposite effect. On the lower pressure surface, an increase in pressure contributes positively to lift, while a pressure reduction tends to decrease it.

For the horizontal porous airfoil, the pressure differences are mainly concentrated around the chordwise pore ends. Because both ends of each pore lie on the same airfoil surface, the internal flow redistributes pressure between different chordwise locations. The combined pressure changes on the upper and lower surfaces increase the pressure difference across the airfoil, consistent with the higher lift coefficient observed for this configuration.

The vertical porous airfoil produces a different pattern because each pore connects the pressure surface to the suction surface. Flow is therefore driven from the higher-pressure region towards the lower-pressure region, producing local pressure changes at both pore ends. At the same time, this cross-surface flow partially reduces the pressure difference between the two surfaces, which is consistent with the lower lift coefficient obtained for the vertical configuration.

The horizontal-vertical porous airfoil contains features associated with both pore arrangements. The horizontal pores redistribute pressure along the airfoil surfaces, while the vertical pores transfer flow between the pressure and suction surfaces. Although the two pore systems remain physically separate and hydraulically independent, their aerodynamic effects interact through the external pressure field. Consequently, the pressure equalisation associated with the vertical pores partly counteracts the pressure redistribution produced by the horizontal pores, giving an overall aerodynamic behaviour closer to that of the impermeable airfoil.

3.4.2. Velocity field characteristics

The velocity field is presented as the percentage difference relative to the impermeable-airfoil solution,

$$\Delta|\mathbf{V}|_{\%} = \frac{|\mathbf{V}|_{\text{porous}} - |\mathbf{V}|_{\text{solid}}}{|\mathbf{V}|_{\text{solid}}} \times 100. \quad (24)$$

This comparison isolates the changes introduced by the pores from the dominant background flow around the airfoil. The largest velocity differences occur close to the pore ends, where the imposed surface-normal flow locally perturbs the external velocity field. Adjacent regions of acceleration and deceleration are visible around several pores and correspond closely to the pressure patterns shown in Figure. 5.

The velocity and pressure fields show the expected inverse relationship described by Bernoulli's equation. Regions where the flow accelerates are generally associated with a reduction in static pressure, whereas regions of lower velocity correspond to higher static pressure. The velocity contours therefore help explain how the pore-induced changes in the local flow field lead to the observed pressure

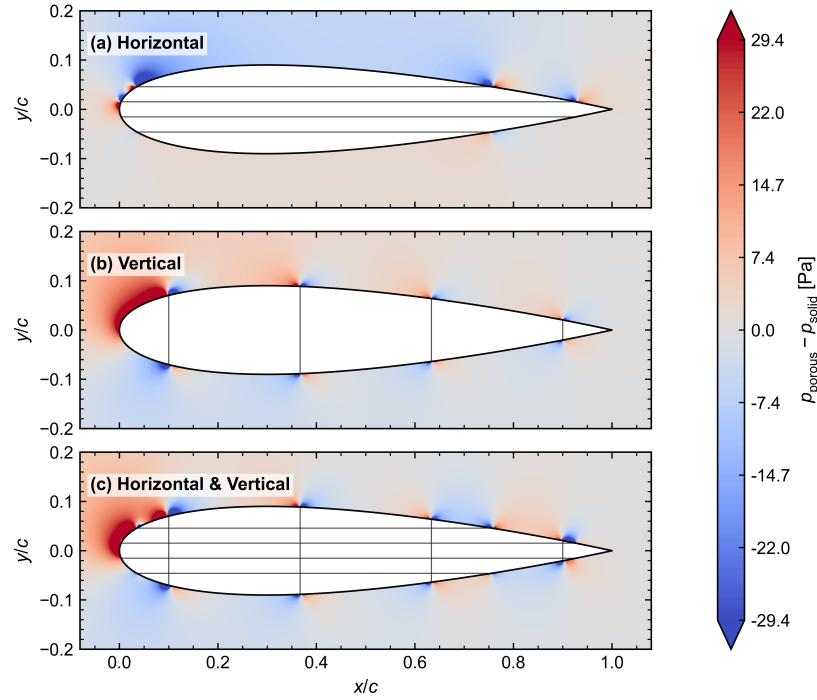


Figure 5: Pressure-field difference between the porous and impermeable airfoils at $\alpha = 10^\circ$ and $H = 9$ mm for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Positive values indicate an increase in pressure relative to the impermeable airfoil, while negative values indicate a reduction.

redistribution and, consequently, changes in the aerodynamic forces and pitching moment.

For the horizontal porous airfoil, alternating regions of acceleration and deceleration are concentrated around the chordwise pore ends. Since both ends of each pore lie on the same airfoil surface, the internal flow redistributes the external velocity between different chordwise locations. The combined changes on the suction and pressure surfaces strengthen the pressure difference across the airfoil, consistent with the increase in lift and the larger change in pitching moment observed for this configuration.

The vertical porous airfoil shows a different velocity pattern. Flow between the pressure and suction surfaces changes the external velocity on both sides of the airfoil and reduces part of the velocity contrast between them. This behaviour is consistent with the partial pressure equalisation discussed previously and the corresponding reduction in lift. A broader region of velocity difference is also visible near the leading edge, where the external flow experiences strong pressure and velocity gradients.

The horizontal–vertical porous airfoil contains features associated with both pore arrangements. The horizontal pores produce local chordwise acceleration and deceleration, while the vertical pores redistribute flow between the pressure and suction surfaces. Although the two pore arrangements remain physically separate and hydraulically independent, their aerodynamic effects interact through the external flow field. The resulting velocity distribution therefore reflects the combined influence of both mechanisms

rather than a simple superposition of the individual configurations.

Overall, the velocity contours support the trends observed in the pressure field and aerodynamic coefficients. The horizontal pores primarily redistribute the flow along the airfoil surfaces, whereas the vertical pores transfer flow between the pressure and suction surfaces. These different flow mechanisms contribute to the contrasting lift and pitching-moment behaviour of the three porous-airfoil configurations.

4. Conclusion

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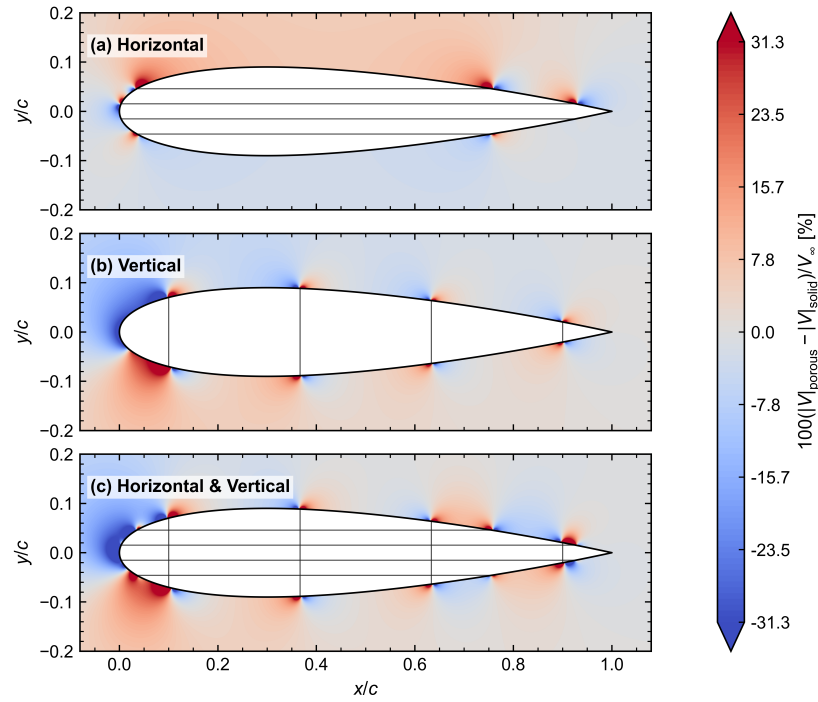


Figure 6: Percentage difference in velocity magnitude between the porous and impermeable airfoils at $\alpha = 10^\circ$ and $H = 9$ mm for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Positive values indicate an increase in velocity magnitude, while negative values indicate a reduction relative to the impermeable airfoil.

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